

Theory of Computation

Submission link: <https://forms.gle/9QfcWiqewGmCcvG36>

Exercise 5: (Regular Grammar)

1. Give the right-linear grammar for the language
 $L = \{1^n 0^m : n \geq 1 \text{ and } m \geq 2\}$.

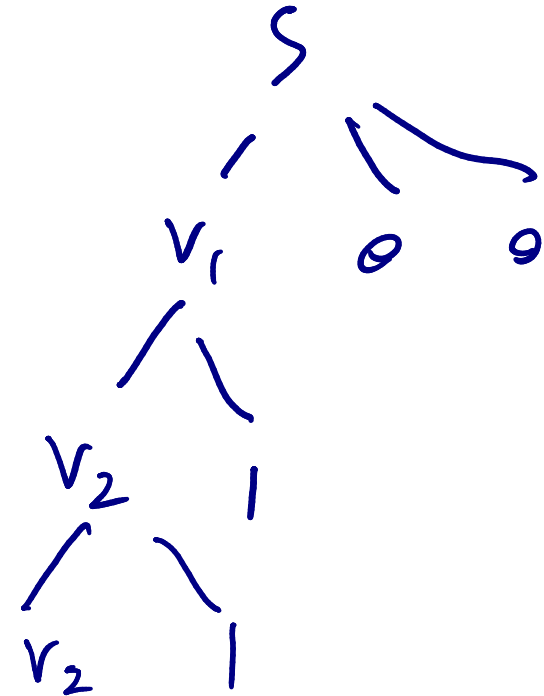
$S \Rightarrow 1A$
 $A \Rightarrow 1A \mid 00B$
 $B \Rightarrow 0B \mid \lambda$

$S \Rightarrow 1A$
 $A \Rightarrow 1A \mid 0B$
 $B \Rightarrow 0B \mid 0$



2. Give the left-linear grammar for the language L in problem 1.

$S \rightarrow B00$
 $B \rightarrow B0 \mid A1$
 $A \rightarrow A1 \mid \lambda$



*3. Draw NFA for the following grammar G.

$G: A \rightarrow xyB,$

$B \rightarrow yxC,$

$C \rightarrow xB \mid yy$

